PROOFLINE 2.0

Evidence-First AI Governance, Verifiable Quorum & Cryptographic Approval Platform

Round 3 Final Product Submission Document

Item / Metric	Submission Details
Live Target Link	https://proofline-ruby-three.vercel.app/
Source Code Repository	https://github.com/AlamNawazSha/proofline
Team Identifier	syedsuhailahamed.m26
Submission Round	Round 3 — The Finish (Final Deliverable)
Automated Test Suite	**100 passed in 21.68s** (100% clean test execution across 4 test suites)
Frontend Production Build	Next.js 15.5.28, React 19, TypeScript — 0 errors, compiled in 6.8s
Core Technology Stack	FastAPI, Python 3.13, Ed25519 Cryptography, PyPDF, SQLite WAL / Supabase Adapter, Next.js 15

1. The Three Improvements Delivered (Round 2 Commitments)

In docs/ROUND2_SUBMISSION.md and docs/ROUND3.md, Proofline explicitly named three substantial improvements. All three have been fully built, verified, and integrated into both the backend API and frontend UI:

Improvement 1: Evidence Lifecycle — Automatic Re-Review Queue

- **Baseline in Round 2:** Managers saw an ephemeral preview of affected documents; manual inspection was required.
- **Delivered in Round 3:** Persistent reReviewQueue state with automatic cascade analysis. Retiring an active source, editing a confirmed rule, or detecting expired sources automatically crawls downstream dependencies, flags affected documents, and queues assigned reviewer tasks.
- **Auditor Resolution Workflow:** Dedicated API endpoints (/api/re-review-queue and /api/re-review-queue/{tid}/resolve) require an explicit justification note before clearing tasks.
- **UI Integration:** Dedicated interactive Re-Review Queue tab in the Approval Center with status badges, assignee labels, and inline resolution drawers.
- **Verification Proof:** Verified in tests/test_round3_improvements.py::test_re_review_queue_lifecycle.

Improvement 2: Measured Detection Quality for AI Review (30-Item Benchmark)

- **Baseline in Round 2:** Deterministic rules were tested, but no frozen evaluation benchmark or published accuracy metrics existed.
- **Delivered in Round 3:** Frozen 30-case evaluation benchmark (BENCHMARK_CASES) covering tone superlatives, unsupported SLA/price claims, prompt injections, expired certificates, ambiguous claims, channel scopes, and true negatives.
- **Measured Metrics:** 100% Accuracy (30/30), 1.000 Precision, 1.000 Recall, 0.00% False Positive Rate, 0.00% False Negative Rate, 100% Quote Validity, and sub-millisecond per-case latency breakdown.
- **UI Integration:** Interactive Adversarial Lab tab enabling live benchmark execution, confusion matrix visualization, and real-time adversarial text testing.
- **Verification Proof:** Verified in tests/test_round3_improvements.py::test_measured_detection_quality_30_case_benchmark.

Improvement 3: Multi-Reviewer Approval Policy, Quorum & Revocation

- **Baseline in Round 2:** Single reviewer approval; no dual quorum for sensitive copy, and no explicit revocation.

• **Delivered in Round 3:** Channel-driven quorum enforcement (public press releases and high-risk documents mandate 2 distinct reviewer signatures). Duplicate reviewer approvals and author self-approval are blocked (403/409).

• **Explicit Revocation:** POST /api/documents/{did}/revoke allows reviewers to revoke an approved release with mandatory justification logging, instantly blocking release export.

• **UI Integration:** Live Quorum status badges (1 of 2 Verified Signatures) and inline Revoke Approval workflows with justification prompts in the Approval Center.

• **Verification Proof:** Verified in tests/test_round3_improvements.py::test_multi_reviewer_quorum_enforcement.

2. Complete Inventory of Features and User Journeys

Governance Module	Key Implemented Capabilities
Content Library & Creator	Multi-format ingestion (.txt, .md, .pdf ≤5 pages, .docx, .csv), SSRF-protected URL ingestion, 7 deterministic rule classes, explicit evidence state badges (SUPPORTED, CONTRADICTED, OUTDATED_SOURCE, INSUFFICIENT_EVIDENCE).
Policy Studio	Lifecycle rule management, quote-verified activation, Natural Language Policy Builder (/api/policies/build-nl), and isolated deterministic policy sandbox (/api/rules/sandbox).
Human Approval Center	Separation of duties (author self-approval blocked with 403), dual-reviewer quorum on high-risk copy, automatic re-review queue for expired/modified evidence, and explicit approval revocation.
Correction Workspace	Interactive diff suggestions with number-preservation checks, evidence-backed compliant rewrites, and author contest rebuttal workflows.
Adversarial Lab	Frozen 30-item evaluation benchmark, confusion matrix metrics (precision, recall, FPR, FNR), latency breakdown, and live adversarial prompt injection testing.
Audit Explorer & Cryptography	Ed25519 digitally signed release receipts, single-character tamper detection, hash-chained audit ledger (SHA256(prev + current)), and full CSV/JSON ledger exports.

3. Technical Architecture and Cryptographic Invariants

Proofline operates on a zero-trust, evidence-grounded architecture where compliance is verified deterministically by code rather than hallucinated by generative models.

Core Cryptographic Invariant:

```
Every release receipt is bound to an asymmetric Ed25519 digital signature over a canonical sorted SHA-256
fingerprint:
Digest = SHA256(CanonicalJSON({ 'context': doc.context, 'text': doc.text, 'visualCheckIds':
doc.visualCheckIds }))
If any single character of document text, declared distribution channel, target audience, or linked visual
asset changes after approval, candidate fingerprint computation fails with CONTENT_INTEGRITY_COMPROMISED,
blocking release export with HTTP 409 Conflict.
```

Separation of Duties & Quorum Invariants:

- **Author Self-Approval Block:** creator(doc) != approver(doc). Enforced on server-side mutate transactions.
- **Dual Quorum Invariant:** High-risk channels mandate |approvers| >= 2 with unique reviewer identities.
- **Fail-Closed Evidence:** Active rules quoting retired or expired sources automatically add high-severity source findings to documents, blocking release readiness.

4. Testing Details & Quality Assurance

Proofline's test suite achieves 100% clean automated execution across 100 distinct unit, integration, and security tests:

Test Suite File	Tests	Key Capabilities Verified
tests/test_core.py	36 passed	Separation of duties, happy journey, source staging, rule deactivation on quote loss, visual asset 8×8 hash matching, WCAG contrast formula, terminology boundaries.
tests/test_upgrades.py	51 passed	Decimal percentages, guarantee synonyms, prompt injection immunity, bearer token authentication, Supabase CAS RPC, SQLite WAL pragmas, rate limiting.
tests/test_proofline2.py	9 passed	Ed25519 passport signatures, risk intelligence matrix, impact cascade, adversarial benchmark runner, hash-chain audit ledger, SSRF protection, NL policy builder.
tests/test_round3_improvements.py	4 passed	Improvement 1 (re-review queue lifecycle), Improvement 2 (frozen 30-case benchmark metrics), Improvement 3 (dual-reviewer quorum & revocation), single-char tamper detection.
TOTAL	**100 passed**	**Execution Time: 21.68s** (Python 3.13 / pytest 9.1.1)

5. Statement of What is Left Undone (Intellectual Honesty)

To uphold rigorous intellectual honesty and transparency for Round 3 judging:

1. **Live External LLM API Provider:** Contextual AI review, prompt injection filtering, and NL policy builders are implemented and tested with mocks. Live OpenAI API execution requires a user-provided OPENAI_API_KEY. Without a key, Proofline fails safely and transparently without generating fabricated AI results.
2. **Enterprise SAML 2.0 / SSO:** Multi-role PBKDF2 authentication with workspace isolation is fully operational. SAML / Okta directory synchronization is architected for Phase 1 of the enterprise roadmap.
3. **Distributed Background Tasks (Celery/Redis):** Re-review queue generation currently executes synchronously during atomic state mutations. Enterprise-scale asynchronous worker queues are planned for Phase 2.

6. Product Roadmap and Future Development Plan

• **Phase 1 (Months 1–3): Enterprise Readiness:** SAML 2.0 / Okta SCIM directory sync, continuous regulatory RSS feed ingestion (FDA, SEC, EU AI Act), and Git-style campaign document branching.

• **Phase 2 (Months 4–6): Autonomous Compliance Scaling:** Multi-tenant vector RAG (hybrid BM25 + pgvector) for >100k page corpora, Google Docs/Figma compliance linting sidecar, and AWS CloudHSM signing.

• **Phase 3 (Months 7–12): Continuous Assurance:** SOC-2 Type II continuous evidence attestations and full-page computer vision for multi-modal brochure layout and typography compliance.